The disc

The disc that the seed's own angular momentum would make, and how the gas settles into it

A companion note to *From a primordial fluctuation to an idealised rotation curve*, Sect. 6.5

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Abstract

Part I. In the seed of the main paper every shell rotates rigidly with the specific angular momentum it receives by turnaround, $j \propto M$. If the gas of each mass element keeps its own angular momentum and settles on a circular orbit in the final rotation curve, the disc that forms is fixed completely by the seed. We derive its surface density in closed form for a flat rotation curve, compute it numerically for the P18 curve with its core, and compare it with an exponential disc and with the relation of Mo et al. (1998) used in the main paper. The seed disc is close to exponential between one and four scale lengths, but it has twice the exponential mass inside half a scale length and a sharp edge at $R_{\max} = 3\sqrt{2}\lambda GM/v \simeq 4.3\,R_d$. Its scale length is 1.06 to 1.4 times the Mo–Mao–White value, and it depends only weakly on the spin. This lowers the inferred retained fraction of angular momentum f_j and leaves the failed spin test as it is. *Part II.* How does a gas parcel arriving on a tilted orbit join the disc? A free particle cannot: its orbit keeps its plane, and with its angular momentum conserved it cannot spiral in. We set up the simplest model with dissipation and a preferred axis: a drag towards gas that rotates about z, at a rate κ times the orbital frequency, with the flow rotating at a fraction f of the circular speed. Where the curve is flat, the parcel spirals in on an exact logarithmic spiral ($\sin p = \kappa$ for $f = 0$); for weak drag $\tan p = \kappa(1-f)/(1 + \mathrm{d}\ln v_c/\mathrm{d}\ln R)$, so the pitch halves in the core of P18. The tilt decays as $\tan(i/2) \propto r^{f/[2(1-f)]}$: without a rotating flow the orbit never settles into the disc. On the sphere of directions the path is an inclined great circle whose tilt decays, not a loxodrome; the loxodrome survives in geometry, since its stereographic projection is an exact logarithmic spiral. The model describes the paths of gas parcels, not spiral arms.

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Part I
The disc of the seed

1 Motivation

The main paper tests the energy closure with discs: if the stellar disc keeps a fraction f_j of the seed's specific angular momentum, then $R_d/r_c = f_j\lambda y/\sqrt{2}$ (its Sect. 6.5). This relation assumes an exponential disc in a flat rotation curve, for which the specific angular momentum is $j_d = 2vR_d$ (Fall & Efstathiou, 1980; Mo et al., 1998). But the seed carries more information than its mean angular momentum: it fixes how the angular momentum is *distributed* over its mass. Under detailed conservation, each mass element keeping its own j (Mestel, 1963; Fall & Efstathiou, 1980), this distribution fixes the disc. The question of Part I is whether the disc so obtained is exponential, and how its scale length compares with the Mo–Mao–White value. Part II asks how a gas parcel arriving on a tilted orbit joins this disc.

2 Theory

2.1 Angular momentum of the seed

As in the main paper we use units with $G = v = 1$, where v is the flat speed, and measure masses in units of the seed mass $M_{\rm tot}$. A shell enclosing the mass M turns around at $R_{\rm ta} = 2GM/v^2$ and receives by then, through tidal torques, the specific angular momentum

$$j_{\rm sh}(M) = \sqrt{2}\,\lambda\, vR_{\rm ta} = 2\sqrt{2}\,\lambda\,\frac{GM}{v}, \tag{1}$$

chosen so that the total angular momentum, $J = \int_0^M j_{\rm sh}\,{\rm d}M' = \sqrt{2}\lambda GM^2/v$, returns the spin parameter of Peebles (1969), $\lambda = J|E|^{1/2}/(GM^{5/2})$ with $|E| = \frac{1}{2}v^2M$, for every M. Each shell

rotates rigidly about a common axis z. A mass element at polar angle θ has $j_z = \omega R^2 \sin^2\theta$, and averaging $\sin^2\theta$ over the sphere ($\langle\sin^2\theta\rangle = \frac{2}{3}$) gives $j_{\rm sh} = \frac{2}{3}\omega R^2$; hence

$$j_z = \tfrac{3}{2}\, j_{\rm sh}(M)\, \sin^2\theta = 3\sqrt{2}\,\lambda\,\frac{GM}{v}\,(1-\mu^2), \qquad \mu = \cos\theta. \tag{2}$$

Since the shells are equally spaced in M and the mass of a shell is spread uniformly in μ, the seed is a population of mass elements with $M/M_{\rm tot}$ and μ uniformly distributed in $[0,1]$. The largest angular momentum, on the equator of the outermost shell, is $j_{\rm max} = 3\sqrt{2}\lambda G M_{\rm tot}/v$, and the mean is $\langle j_z\rangle = j_{\rm max}\langle M/M_{\rm tot}\rangle\langle 1-\mu^2\rangle = j_{\rm max}\cdot\frac{1}{2}\cdot\frac{2}{3} = \sqrt{2}\lambda G M_{\rm tot}/v$, the specific angular momentum of the seed.

2.2 Detailed conservation and the disc

Let a fraction of the matter be gas that keeps a fraction f_j of its own j_z and settles on the circular orbit with that angular momentum in the final rotation curve $v_c(R)$:

$$R\, v_c(R) = f_j\, j_z. \tag{3}$$

Then the distribution of j_z over the mass becomes the distribution of disc mass over radius. We take the gas to be a fixed fraction of every mass element, so the disc mass M_d is distributed as the seed's j_z.

Flat rotation curve (analytic). With $v_c = v$ the orbit radius is $R = f_j j_z/v$, and with $S = R/R_{\rm max}$,

$$R_{\rm max} = \frac{f_j j_{\rm max}}{v} = 3\sqrt{2}\, f_j \lambda\, \frac{GM_{\rm tot}}{v^2}, \qquad S = \frac{M}{M_{\rm tot}}\,(1-\mu^2). \tag{4}$$

The disc mass inside S is the fraction of the unit square $(M/M_{\rm tot}, \mu)$ where $(M/M_{\rm tot})(1-\mu^2) < S$. For $1-\mu^2 < S$, i.e. $\mu > \sqrt{1-S}$, every M qualifies; otherwise only $M/M_{\rm tot} < S/(1-\mu^2)$. Hence

$$F(S) \equiv \frac{M_d(<R)}{M_d} = \left(1-\sqrt{1-S}\right) + \int_0^{\sqrt{1-S}} \frac{S\,{\rm d}\mu}{1-\mu^2} = 1 - \sqrt{1-S} + S\ {\rm artanh}\,\sqrt{1-S}, \tag{5}$$

using $\int {\rm d}\mu/(1-\mu^2) = {\rm artanh}\,\mu$. Differentiating, the two terms $\pm 1/(2\sqrt{1-S})$ cancel and $F'(S) = {\rm artanh}\,\sqrt{1-S}$, so the surface density, $\Sigma = ({\rm d}M_d/{\rm d}R)/(2\pi R)$, is

$$\Sigma(R) = \frac{M_d}{2\pi R_{\rm max}^2}\,\frac{{\rm artanh}\,\sqrt{1-S}}{S}, \qquad S = \frac{R}{R_{\rm max}} \le 1. \tag{6}$$

Two limits show its character. Near the centre, ${\rm artanh}\,\sqrt{1-S} \simeq \frac{1}{2}\ln(4/S)$, so $\Sigma \propto \ln(4/S)/S$: a central cusp, steeper than any exponential. At the edge, ${\rm artanh}\,\sqrt{1-S} \simeq \sqrt{1-S}$, so $\Sigma \to 0$ as $\sqrt{1-S}$: the disc ends sharply at $R_{\rm max}$, the orbit of the equator of the outermost shell. In between, Eq. (6) is close to exponential (Fig. 1a).

P18 rotation curve (numerical). Inside the core the curve rises, $v_c \simeq (2/\sqrt{3})(v/r_c)R$, so Eq. (3) gives $R \propto j_z^{1/2}$ instead of $R \propto j_z$: low-j material lands further out than in a flat curve, and the central cusp of Eq. (6) is softened. We invert Eq. (3) on a grid for the P18 curve $v_c = v\sqrt{m(x)/x}$, $x = R/r_c$, with $r_c = R_b/y$ and the core ratio y of the main paper (its core equation, Sect. 4.3, with $\Lambda = 0$; $y = 50.5$ for $\lambda = 0.035$), and sample 2×10^6 mass elements from the uniform distribution of $(M/M_{\rm tot}, \mu)$.

Scale length of a non-exponential disc. For the comparison with an exponential disc a scale length must be defined. We use two definitions: an exponential fit to Σ between one and four scale lengths (iterated), as photometry would measure it, and the half-mass radius divided by 1.678, its value for an exponential disc. The relation of Mo et al. (1998) for a flat curve gives $R_d^{\rm MMW} = f_j\langle j_z\rangle/(2v) = f_j\lambda G M_{\rm tot}/(\sqrt{2}v^2)$, i.e. $R_d/r_c = f_j\lambda y/\sqrt{2}$ with $GM_{\rm tot}/v^2 \simeq R_b$.

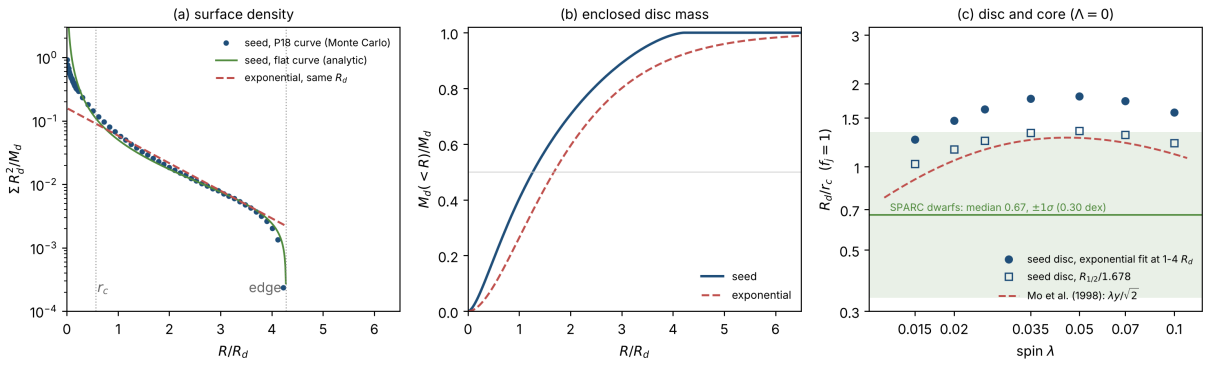


Figure 1: The disc made by the seed's own angular momentum ($f_j = 1$, $\Lambda = 0$). (a) Surface density for $\lambda = 0.035$: Monte Carlo in the P18 curve (points), the analytic flat-curve result, Eq. (6) (solid), and an exponential disc with the same scale length (dashed); dotted lines mark the core radius and the edge $R_{\max}$. (b) Enclosed disc mass, seed against exponential. (c) Ratio of disc scale length to core radius against spin: seed disc with R_d from an exponential fit at 1–4 R_d (filled) and from the half-mass radius (open), and the relation of Mo et al. (1998) used in the main paper (dashed). Green: the median and 1σ spread of the 79 SPARC dwarfs; the observed values require $f_j < 1$.

3 The code

The script `seed_disc.py` (Appendix A) does the following. (i) It solves the core equation of the main paper for y. (ii) It draws mass elements with uniform $M/M_{\rm tot}$ and μ, assigns j_z from Eq. (2), and places each on the circular orbit of Eq. (3) in the P18 curve, by interpolating the monotonic function $Rv_c(R)$. (iii) It computes Σ, the enclosed mass and the two scale lengths, and compares them with the analytic flat-curve result, Eqs. (5) and (6), and with an exponential disc. (iv) It repeats the calculation for spins from 0.015 to 0.1 and makes Fig. 1. Near the centre the closed form of $m(x)$ loses precision through cancellation; the code uses the series $m \simeq \frac{4}{3}x^3$ for $x < 10^{-3}$. It runs in about twenty seconds.

4 Results

For the median spin, $\lambda = 0.035$ ($y = 50.5$, $r_c = 0.020\,R_b$), the script gives the following.

- *Shape.* Between one and four scale lengths the disc is close to exponential (Fig. 1a). Inside $0.5\,R_d$ it holds 19% of its mass, twice the 9% of an exponential disc; the half-mass radius is $1.27\,R_d$ instead of $1.68\,R_d$ (Fig. 1b). The disc ends at $R_{\max} = 0.149\,R_b = 4.3\,R_d$.
- *Effect of the core.* The rising inner curve moves low-j material outward: inside $S = 0.1$ the P18 curve places 15% of the disc mass, the flat curve 23%. Beyond $S \simeq 0.3$ the two agree.
- *Scale length.* The exponential fit gives $R_d = 0.035\,R_b$, $R_d/r_c = 1.76 f_j$, 1.40 times the Mo–Mao–White value $1.26 f_j$; the half-mass definition gives $1.33 f_j$, 1.06 times that value. The difference between the two definitions measures how far the seed disc is from exponential.
- *Dependence on spin.* R_d/r_c changes little with λ (Fig. 1c): $\mathrm{d}\ln(R_d/r_c)/\mathrm{d}\ln\lambda = 0.16$ at $\lambda = 0.035$, with a broad maximum near $\lambda \simeq 0.05$. The reason is the same as in the main paper: a larger spin makes the disc larger but also the core larger (y smaller), and the two nearly cancel.

5 Relevance to the main paper

The retained fraction f_j. The observed median $R_d/r_c = 0.67$ is reproduced with a smaller f_j than with the Mo–Mao–White relation: smaller by the factor 1.06 to 1.4, so the $f_j \simeq 0.77$ of

the main paper becomes 0.55–0.73. This is within the range usually inferred for discs and does not change any conclusion.

The spin test. The test of the main paper rests on the *width* of the distribution of R_d/r_c. A constant factor does not change the width, and the weak dependence on λ found here (0.16) is the same as with the Mo–Mao–White relation. The predicted distribution therefore stays as narrow as before, and the spin test fails as before.

The shape of real discs. Detailed conservation of the seed's angular momentum gives too much mass at the centre and a sharp edge, the well-known angular-momentum distribution problem of discs that form by detailed conservation (Dalcanton et al., 1997; Bullock et al., 2001; van den Bosch et al., 2001). Late-type dwarfs are bulgeless, so the central excess must have been removed or redistributed; preferential removal of low-angular-momentum gas by feedback is the usual explanation (Governato et al., 2010; Brook et al., 2011). The edge near $4.3\,R_d$ is reminiscent of the truncations and breaks observed in disc profiles at three to five scale lengths (van der Kruit, 1979; Pohlen & Trujillo, 2006), but a dissipational disc need not keep the sharp edge of Eq. (6).

Part II
How the gas settles into the disc

6 From the loxodrome picture to a model

The idea to be modelled is this: matter collapsing in a rotating protogalaxy winds around a shrinking sphere along a spiral, and when the system flattens, the spiral becomes a logarithmic spiral in the disc. Two simple facts show what a model must contain.

A free particle does not wind around a sphere. In a spherical potential the force is radial, so the torque $\mathbf{r}\times\mathbf{F}$ vanishes and the angular momentum *vector* $\mathbf{L}$ is conserved. The particle moves in the fixed plane perpendicular to $\mathbf{L}$, and its direction $\hat{\mathbf{r}}$ moves on a great circle of the sphere of directions, never on a loxodrome or any other spiral.

A free particle does not spiral in steadily. With $j = rv_\varphi$ conserved, the azimuthal speed grows as $1/r$ when r shrinks, and the particle bounces back from its centrifugal barrier. Even if one imposes a steady inflow, $\dot r = -kr$, the conserved j gives $\dot\varphi = j/r^2$ and hence $\mathrm{d}r/\mathrm{d}\varphi = -(k/j)r^3$, i.e. $r \propto \varphi^{-1/2}$: a lituus, not a logarithmic spiral.

Both facts point to the same missing ingredient: *dissipation*. (Part I assumed it: there each gas element simply settles on the circular orbit of its own angular momentum.) Gas can lose angular momentum and energy in collisions between streams, in shocks and through viscosity; dark matter cannot. The model is therefore a model for the gas of the seed, which in the paper settles with its angular momentum into the disc (Sect. 6.5). A second ingredient is needed to make the disc *one* plane: a preferred axis. In the model it is the rotation of the gas that has already settled.

7 The geometry of constant angles

7.1 The logarithmic spiral

In plane polar coordinates (r,φ) a curve crosses the circle $r=\text{const}$ at the *pitch angle* p given by

$$\tan p = -\frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}\varphi} = -\frac{v_r}{v_\varphi}, \tag{7}$$

where $v_r = \dot{r}$ and $v_\varphi = r\dot{\varphi}$ for a particle moving along the curve. A constant pitch gives $\ln r = -\varphi \tan p + \text{const}$, the *logarithmic spiral* $r = r_0 e^{-\varphi \tan p}$. Kinematically: *a particle moves on a logarithmic spiral if and only if the ratio of its radial to its azimuthal velocity is constant.*

7.2 The loxodrome

On a sphere with latitude ψ and longitude λ, a loxodrome crosses all meridians at the same angle α. A step along it has meridional length $\mathrm{d}\psi$ and zonal length $\cos\psi\,\mathrm{d}\lambda$ (on the unit sphere), so

$$\tan\alpha = \frac{\cos\psi\,\mathrm{d}\lambda}{\mathrm{d}\psi} = \frac{v_\lambda}{v_\psi}, \qquad \lambda - \lambda_0 = \tan\alpha\ \ln\tan\Big(\frac{\pi}{4} + \frac{\psi}{2}\Big). \tag{8}$$

Again a constant ratio, now of the zonal to the meridional velocity. The quantity $y = \ln\tan(\pi/4 + \psi/2)$ is the ordinate of Mercator's map, on which loxodromes are straight lines.

7.3 Stereographic projection turns loxodromes into logarithmic spirals

Project the unit sphere from its south pole onto the equatorial plane. A point at latitude ψ and longitude λ lands at polar angle λ and radius

$$\rho = \frac{\cos\psi}{1+\sin\psi} = \tan\Big(\frac{\pi}{4} - \frac{\psi}{2}\Big) = e^{-y}. \tag{9}$$

On a loxodrome $y = (\lambda - \lambda_0)\cot\alpha$, so its image is

$$\rho = e^{-(\lambda-\lambda_0)\cot\alpha}\ : \tag{10}$$

an exact logarithmic spiral, with pitch $p = 90° - \alpha$. The reason is that stereographic projection is *conformal*: it preserves angles. It maps meridians onto radial lines, and a curve that crosses all meridians at the angle α is mapped onto a curve that crosses all radial lines at the same angle (Fig. 2).

This is the precise sense in which "a spiral on a sphere becomes a logarithmic spiral in the plane". It is a statement of geometry, however, not of dynamics. Moreover, a loxodrome runs from pole to pole and never settles onto the equator; a particle that is to end up in a disc must do something else. The rest of Part II asks what.

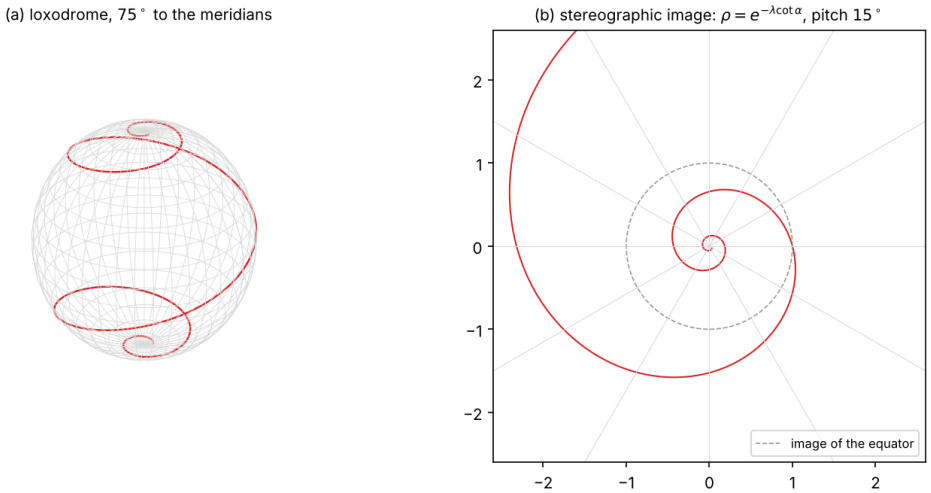


Figure 2: (a) A loxodrome crossing the meridians at 75°. (b) Its stereographic projection from the south pole: the logarithmic spiral $\rho = e^{-\lambda\cot\alpha}$, with pitch 15°. Grey lines: images of meridians (radial lines) and of the equator (unit circle). Script: `code/spiral_figs.py`.

8 The model

A gas parcel moves in the potential Φ of the galaxy, here P18 with $\mathrm{d}\Phi/\mathrm{d}r = GM(r)/r^2$, and feels a drag towards the local mean flow of the gas:

$$\ddot{\mathbf{r}} = -\nabla\Phi - \gamma\big(\dot{\mathbf{r}} - \mathbf{u}\big), \qquad \mathbf{u} = f\,\Omega(r)\,\hat{\mathbf{z}} \times \mathbf{r}, \qquad \gamma = \kappa\,\Omega(r), \tag{11}$$

with $\Omega = v_c/r$ the circular angular frequency. The flow $\mathbf{u}$ rotates about the z-axis with angular speed $f\Omega$: in the equatorial plane at a fraction f of the circular speed ($0 \le f < 1$; gas supported partly by pressure, or still assembling, rotates more slowly than circular). The drag rate is a fixed fraction κ of the orbital frequency, so that it acts with the same strength per orbit at every radius. In the flat part of the curve this makes the model *scale-free*: no length scale is introduced, which is why logarithmic spirals appear.

The parcel's velocity has three components, radial v_r, meridional v_ψ (towards or away from the poles) and azimuthal v_φ (about z). The model couples them as follows.

- The radial motion is pulled by gravity and damped by the drag.
- The angular momentum changes by the torque of the drag,

$$\dot{\mathbf{L}} = -\gamma\,\mathbf{L} + \gamma f\Omega\,\big(r^2\hat{\mathbf{z}} - z\,\mathbf{r}\big), \tag{12}$$

  where the first term shrinks $\mathbf{L}$ without turning it, and the second, from the rotating flow, adds angular momentum about z and turns $\mathbf{L}$ towards $\hat{\mathbf{z}}$.

Equation (12) contains the first result. *Without a rotating flow, $f = 0$, the torque is parallel to $\mathbf{L}$: the orbit loses angular momentum and spirals in, but in its own, tilted plane.* A drag towards rest cannot make a disc. Only friction against matter that already rotates about an axis can bring orbits into one plane.

9 Analytic results

9.1 The flat part: an exact logarithmic spiral

Where $v_c = v$, look for a steady inflow in the equatorial plane with constant velocity ratios, $v_r = -Av$ and $v_\varphi = Bv$. Along such a path v_r and v_φ do not change, and the equations of motion in polar coordinates,

$$\dot{v}_r - \frac{v_\varphi^2}{r} = -\frac{v^2}{r} - \gamma v_r, \qquad \dot{v}_\varphi + \frac{v_r v_\varphi}{r} = -\gamma\,(v_\varphi - fv), \tag{13}$$

with $\gamma = \kappa v/r$, reduce to two algebraic conditions:

$$B^2 = 1 - \kappa A, \qquad AB = \kappa\,(B - f). \tag{14}$$

Eliminating A,

$$B^3 + (\kappa^2 - 1)\,B = \kappa^2 f, \qquad \tan p = \frac{A}{B} = \frac{\kappa\,(B-f)}{B^2}. \tag{15}$$

Since the ratio v_r/v_φ is constant, the path is an exact logarithmic spiral (Eq. 7). For $f = 0$, $A = \kappa$ and $B = \sqrt{1-\kappa^2}$, so that

$$\sin p = \kappa \qquad (f = 0). \tag{16}$$

Two remarks. The parcel moves at slightly less than circular speed ($B < 1$), since the drag holds it back. And the inflow speed Av is a constant fraction of v: the parcel drifts inwards by the same factor per turn at every radius, which is the defining property of a logarithmic spiral. Values are given in Table 1.

Table 1: Steady inflow in the flat part, Eq. (15), and pitch in the solid-body core, Eq. (19); q is the decay rate of the tilt per radian, Eq. (20).

κ	f	B	A	pitch, flat part	q	pitch, core
0.1	0	0.995	0.100	5.7°	0	2.9°
0.1	0.5	0.998	0.050	2.9°	0.025	1.4°
0.2	0	0.980	0.200	11.5°	0	5.7°
0.2	0.5	0.990	0.099	5.7°	0.051	2.9°
0.3	0.5	0.978	0.147	8.5°	0.078	4.3°
0.5	0.5	0.940	0.234	14.0°	0.142	7.1°
0.5	0.8	0.977	0.091	5.3°	0.210	2.8°

9.2 Any rotation curve: the pitch follows the slope of the curve

For weak drag, $\kappa \ll 1$, the parcel stays close to a circular orbit, and its inflow is set by its angular-momentum budget. On a near-circular orbit $j = r v_c(r)$, so moving inwards by $\mathrm{d}r$ costs

$$\frac{\mathrm{d}j}{\mathrm{d}r} = v_c + r\frac{\mathrm{d}v_c}{\mathrm{d}r} = v_c\,(1+\beta_s), \qquad \beta_s \equiv \frac{\mathrm{d}\ln v_c}{\mathrm{d}\ln R}, \tag{17}$$

while the drag removes angular momentum at the rate $\dot{j} = -\gamma(v_c - f v_c)\,r = -\kappa(1-f)\,v_c^2$. The inflow speed is $v_r = \dot{j}/(\mathrm{d}j/\mathrm{d}r) = -\kappa(1-f)\,v_c/(1+\beta_s)$, and hence

$$\boxed{\tan p = \frac{\kappa\,(1-f)}{1 + \mathrm{d}\ln v_c/\mathrm{d}\ln R}} \tag{18}$$

In the flat part, $\beta_s = 0$, this is the small-κ limit of Eq. (15). In a solid-body core, $\beta_s = 1$, it is half as large: there each step inwards costs twice as much angular momentum, because $j = \Omega r^2$ falls as r^2. For P18, β_s falls from 1 at the centre to 0.54 at r_c, 0.27 at $2r_c$ and 0 at the maximum of the curve, $5.5\,r_c$, so that the inflow paths open up from the core outwards (Fig. 3e).

In the harmonic core itself the model is linear and can be solved exactly. With $\zeta = x + iy$ in the equatorial plane and $\Omega = \omega = (2/\sqrt{3})\,v/r_c$ constant, Eq. (11) becomes $\ddot{\zeta} = -\omega^2\zeta - \kappa\omega(\dot{\zeta} - if\omega\zeta)$, whose solutions are $\zeta \propto e^{st}$ with

$$s^2 + \kappa\omega\, s + \omega^2(1 - i\kappa f) = 0, \qquad \tan p = -\frac{\mathrm{Re}\, s}{\mathrm{Im}\, s}, \tag{19}$$

again an exact logarithmic spiral, since $\zeta \propto e^{(\mathrm{Re}\,s)t}e^{i(\mathrm{Im}\,s)t}$. To first order in κ, $s = i\omega - \frac{1}{2}\kappa\omega(1-f)$, which is Eq. (18) with $\beta_s = 1$.

9.3 Settling into the disc

Now let the orbit be tilted by the angle i against the equatorial plane. Average the torque (12) over one revolution on a circle of radius r in the tilted plane. With unit vectors $\mathbf{e}_1$ along the line of nodes and $\mathbf{e}_2$ in the plane, $\hat{\mathbf{z}}\cdot\mathbf{e}_2 = \sin i$, one has $z = r\sin i\sin\phi'$ and $\langle z\mathbf{r}\rangle = \frac{1}{2}r^2\sin i\,\mathbf{e}_2$, where ϕ' is the angle in the orbital plane. The part of the averaged torque perpendicular to $\mathbf{L}$ is therefore $\frac{1}{2}\gamma f\Omega r^2\sin i\;\mathbf{e}_2$, which turns $\mathbf{L}$ towards $\hat{\mathbf{z}}$ at the rate $\mathrm{d}i/\mathrm{d}t = -\gamma f\Omega r^2\sin i/(2j)$. With $j = Bvr$ and the orbital rate $\mathrm{d}\phi'/\mathrm{d}t = Bv/r$,

$$\frac{\mathrm{d}i}{\mathrm{d}\varphi} = -q\sin i, \qquad q = \frac{\kappa f}{2B^2}, \qquad \text{so} \qquad \tan\frac{i}{2} = \tan\frac{i_0}{2}\,e^{-q\varphi}. \tag{20}$$

The tilt decays exponentially with the angle turned, but only if $f > 0$. Dividing by the inflow per radian, $\mathrm{d}\ln r/\mathrm{d}\varphi = -\tan p$, gives the tilt as a function of radius,

$$\tan\frac{i}{2} \propto r^{\,q/\tan p} = r^{\,f/[2(B-f)]} \simeq r^{\,f/[2(1-f)]}, \tag{21}$$

independent of κ. For $f = 0.5$ the tilt falls as $r^{1/2}$: the parcel must move in by a factor of four to halve its tilt. For $f = 0.9$ it falls as $r^{4.5}$: a flow that nearly co-rotates flattens the orbit long before the parcel has moved far in. The thinness of a disc thus measures how close to co-rotation the gas was when it settled. (In the core the pitch halves, so the exponent of Eq. 21 doubles.)

9.4 The path on the sphere

What does the parcel's direction $\hat{\mathbf{r}}$ do? On a tilted orbit the latitude oscillates, $\sin\psi = \sin i \, \sin\phi'$, once per revolution between $-i$ and $+i$. As i decays, the oscillation shrinks, and the path of $\hat{\mathbf{r}}$ on the sphere of directions is a great circle that tilts down onto the equator turn by turn (Fig. 3c). Its heading against the meridians is not constant, as for a loxodrome, but turns towards the east: the path winds onto the equator instead of running to a pole. Combined with the shrinking radius, the parcel moves on a sequence of ever smaller spheres along ever flatter great circles, and ends on the logarithmic spiral of Sect. 9.1 in the plane. This is the dynamical version of the picture we started from: a spiral on a sphere that becomes a logarithmic spiral in a disc.

10 Numerical check

The equations (11) were integrated for parcels starting at $r = 60\,r_c$ on the steady inflow of the flat part, but in a plane tilted by 40° (`code/spiral_model.py`). Figure 3 and Table 2 show the results.

Table 2: Predicted and measured pitch angle (in the orbital plane, averaged over all passages through the given radii) and tilt decay rate.

κ	f	pitch at 8–12 r_c		pitch at 0.2–0.5 r_c		$\mathrm{d}\ln\tan(i/2)/\mathrm{d}\varphi$	
		predicted	measured	predicted	measured	predicted	measured
0.2	0.5	5.7°	6.1°	3.1°	3.1°	−0.051	−0.050
0.3	0.5	8.5°	9.1°	4.6°	4.7°	−0.078	−0.081
0.2	0	11.5°	11.7°	6.2°	6.3°	0	0

The predictions hold to a few per cent. The pitch in the flat part is slightly larger than predicted while the orbit is still tilted: a tilted orbit sees the rotating flow at an angle, so the effective f is smaller (roughly $f\cos i$), and the inflow faster. Once the orbit has settled, the measured pitch follows Eq. (18) through the transition from the flat part to the core. For $f = 0$ the tilt stays exactly at 40°, as Eq. (12) requires.

11 What the model means, and what it does not

- *It is a model for gas.* Dark matter has no drag; its orbits keep their planes and their angular momenta (companion note *Halting and settling*). The model describes how a gas parcel of the seed, arriving on a tilted, nearly circular orbit after its shock at the front, would join the disc that the gas already in place has formed (paper, Sect. 6.5).

- *Logarithmic spirals come from scale freedom.* The paths are exactly logarithmic where the rotation curve is flat or rising linearly, because there the model has no length scale; in between, their pitch follows the slope of the curve, Eq. (18). For P18 the pitch of the inflow halves from the flat part to the core.

- *A disc needs a rotating flow.* Without it ($f = 0$) drag lets matter spiral in, but each parcel in its own plane. The rate of flattening per radius depends only on f, Eq. (21).

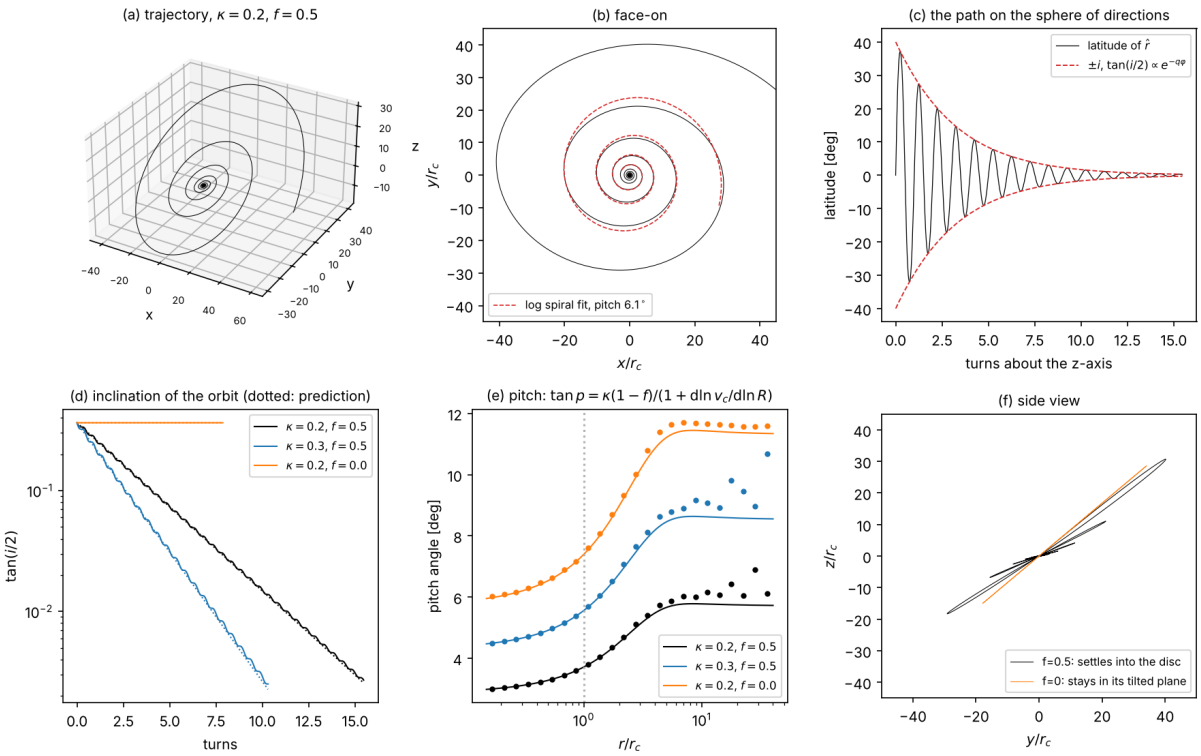


Figure 3: The model in the P18 potential (units $G = v = r_c = 1$). (a) Trajectory for $\kappa = 0.2$, $f = 0.5$, starting at $60\,r_c$ in a plane tilted by 40°. (b) Face-on view with a logarithmic spiral fitted at 3–30 r_c (pitch 6.1° in projection, against 5.7° predicted for a settled orbit). (c) Latitude of the direction $\hat{\mathbf{r}}$ against the number of turns: oscillations between $\pm i$, with the envelope of Eq. (20). (d) Tilt of the orbit, $\tan(i/2)$, for three models; dotted: Eq. (20). (e) Pitch angle against radius: points from the integrations, lines from Eq. (18). (f) Side view along the line of nodes: with $f = 0.5$ the orbit settles into the disc; with $f = 0$ it spirals in within its tilted plane. Script: `code/spiral_figs.py`.

- *The paths are not spiral arms.* The arms of disc galaxies are patterns, density waves or transient swing-amplified features, through which the gas and stars move; material arms would wind up within a few rotations. The model's spirals are the paths of individual parcels, typically with pitch angles of a few degrees for $\kappa \lesssim 0.3$, much tighter than typical arms (10–30°). Their pitch also *grows* with the shear of the rotation curve (larger in the flat part than in the core), while the arms of observed galaxies are reported to be *tighter* where the shear is larger (Seigar et al. 2005, 2006); the two are different things.

- *The parameters are not derived.* κ and f summarise collisions, shocks and viscosity. A hydrodynamical simulation of the seed's gas would fix them; the model only shows what follows once they are given.

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A Code

Part I: `code/seed_disc.py` (listed below), output `code/seed_disc_output.txt`. Part II: `code/spiral_model.p` (analytic predictions and integrations; output `code/spiral_model_out.txt`), `code/spiral_figs.py` (the two figures of Part II; it needs `spiral_runs.pkl`, written by `spiral_model.py`) and `code/p18core.py` (the P18 functions). Run the scripts from `code/`.

```
"""The disc that the seed's own angular momentum would make.

Seed of pf2irc: shells of mass M with one specific energy -v^2/2 and spin lambda.
Each shell rotates rigidly with the specific angular momentum it has at turnaround,
    j_shell(M) = 2 sqrt2 lambda G M / v ,
so a mass element at polar angle theta has j_z = (3/2) j_shell sin^2(theta).
Detailed angular-momentum conservation: the gas element settles on the circular orbit
    R v_c(R) = f_j j_z
of the P18 rotation curve v_c = v sqrt(m(x)/x), x = R/r_c, with r_c = R_b / y.

Units: G = v = 1, total seed mass M_tot = 1, so R_b(1 + 1/4y) = 1.
Outputs: numbers printed, figure fig_seed_disc.png.
Run: python3 seed_disc.py (numpy, scipy, matplotlib; ~20 s)
"""
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import lambertw

CE = -0.91737
def y_of(lam, ell=0.0): # core ratio of pf2irc, Eq. (lambert)
    A = 3*lam**2 + 2*ell/3; B = 0.25 - CE - 0.75*lam**2
    return lambertw(4*A*np.exp(4*B)).real/(4*A)
def m(x):
    x = np.asarray(x, float)
    full = x*(1-np.exp(-x))**2 + 0.25*(1-(1+2*x+2*x*x)*np.exp(-2*x))
    return np.where(x < 1e-3, 4*x**3/3, full) # series near the centre avoids round-off
def vc(R, rc): return np.sqrt(m(R/rc)*rc/R) # v_c / v

def F_flat(S): # disc mass fraction inside S = R/R_max for a flat curve (analytic)
    S = np.clip(S, 1e-12, 1); u = np.sqrt(1-S)
    return 1 - u + S*np.arctanh(np.minimum(u, 1-1e-15))
def Sigma_flat(S): # surface density in units of M_d / R_max^2 (analytic)
    S = np.clip(S, 1e-12, 1-1e-15); return np.arctanh(np.sqrt(1-S))/(2*np.pi*S)

def disc(lam=0.035, y=None, ell=0.0, fj=1.0, N=2_000_000, seed=1):
    """Monte Carlo disc radii (units of R_b) for the P18 curve; returns R, R_b, r_c, R_max."""
    y = y_of(lam, ell) if y is None else y
    Rb = 1/(1+1/(4*y)); rc = Rb/y
    rng = np.random.default_rng(seed)
    M = rng.uniform(0, 1, N); mu = rng.uniform(-1, 1, N)
    jz = fj*1.5*2*np.sqrt(2)*lam*M*(1-mu**2)
    Rg = np.logspace(-7, 1, 6000); Jg = Rg*vc(Rg, rc) # invert R v_c(R) = j_z
```

```
    R = np.interp(jz, Jg, Rg)
    Rmax = fj*3*np.sqrt(2)*lam # j_max / v
    return R/Rb, Rb, rc/Rb, Rmax/Rb, y

def exp_fit(R, lo=1.0, hi=4.0, Rd0=None):
    """Exponential scale length from the surface density between lo and hi times R_d (iterated)."""
    Rd = np.mean(R)/2 if Rd0 is None else Rd0
    for _ in range(20):
        e = np.linspace(lo*Rd, hi*Rd, 25); h, _ = np.histogram(R, e)
        S = h/(np.pi*(e[1:]**2-e[:-1]**2)); Rm = 0.5*(e[1:]+e[:-1]); ok = S > 0
        Rd = -1/np.polyfit(Rm[ok], np.log(S[ok]), 1)[0]
    return Rd

if __name__ == "__main__":
    lam = 0.035
    R, Rb, rc, Rmax, y = disc(lam)
    Rd = exp_fit(R); Rd_half = np.median(R)/1.678 # two definitions of R_d for a non-exponential disc
    jmean = np.sqrt(2)*lam # <j_z> = sqrt2 lam G M_tot / v (per M_tot)
    Rd_mmw = jmean/2*(1+1/(4*y)) # exponential disc in a flat curve: j = 2 v R_d (units of R_b)
    print(f"lambda = {lam}, y = {y:.1f}, r_c = {rc:.4f} R_b, R_max = {Rmax:.4f} R_b")
    print(f"exponential fit (1-4 R_d): R_d = {Rd:.4f} R_b = {Rd/Rmax:.3f} R_max; R_d/r_c = {Rd/rc:.2f}")
    print(f"Mo-Mao-White (j_d = 2 v R_d): R_d = {Rd_mmw:.4f} R_b; R_d/r_c = {Rd_mmw/rc:.2f}; ratio {Rd/
        Rd_mmw:.3f}")
    print(f"half-mass radius: {np.median(R)/Rd:.3f} R_d (exponential 1.678); edge R_max = {Rmax/Rd:.2f} R_d"
        )
    print(f"R_d from the half-mass radius (R_half/1.678) = {Rd_half:.4f} R_b; R_d/r_c = {Rd_half/rc:.2f};
        ratio to MMW {Rd_half/Rd_mmw:.3f}")
    e = np.array([0, 0.1, 0.25, 0.5])*Rd
    inner = np.histogram(R, e)[0]/len(R)
    from math import exp
    expo = [ (1-(1+b)*np.exp(-b)) - (1-(1+a)*np.exp(-a)) for a, b in zip(e[:-1]/Rd, e[1:]/Rd)]
    print("mass in R/R_d = 0-0.1, 0.1-0.25, 0.25-0.5: seed", inner.round(4), " exponential", np.round(expo,
        4))
    # flat-curve analytic check
    S = np.sort(R)/Rmax; Fmc = np.arange(1, len(S)+1)/len(S); idx = np.searchsorted(S, [0.1, 0.3, 0.6])
    print("analytic flat-curve F(S) at S=0.1,0.3,0.6:", F_flat(np.array([0.1, 0.3, 0.6])).round(4),
        " Monte Carlo (P18 curve):", Fmc[idx].round(4))
    # spin dependence of R_d/r_c
    lams = np.array([0.015, 0.02, 0.025, 0.035, 0.05, 0.07, 0.1])
    q_seed, q_half, q_mmw = [], [], []
    for l_ in lams:
        R_, Rb_, rc_, Rm_, y_ = disc(l_, N=600_000)
        q_seed.append(exp_fit(R_)/rc_); q_half.append(np.median(R_)/1.678/rc_)
        q_mmw.append(np.sqrt(2)*l_/2*(1+1/(4*y_))/rc_)
    q_seed, q_half, q_mmw = np.array(q_seed), np.array(q_half), np.array(q_mmw)
    print("R_d/r_c (seed fit 1-4 R_d / seed half-mass / MMW):",
        " ".join(f"{a}:{b:.2f}/{d:.2f}/{c:.2f}" for a, b, d, c in zip(lams, q_seed, q_half, q_mmw)))
    print("d ln(R_d/r_c)/d ln lambda (seed, around 0.035):",
        np.round(np.gradient(np.log(q_seed), np.log(lams))[3], 3))

    # ------------------------------------------------------------------ figure
    C1, C2, C3 = "#1f4e79", "#c0504d", "#4f8a3c"
    fig, ax = plt.subplots(1, 3, figsize=(12.5, 3.9))
    # (a) surface density
    edges = np.concatenate([np.linspace(0, 0.2, 21), np.linspace(0.25, 6.5, 60)])*Rd
    h, _ = np.histogram(R, edges); Rm = 0.5*(edges[1:]+edges[:-1])
    Sig = h/len(R)/(np.pi*(edges[1:]**2-edges[:-1]**2))
    x = Rm/Rd; ok = Sig > 0
    ax[0].semilogy(x[ok], Sig[ok]*Rd**2, "o", ms=3, c=C1, label="seed, P18 curve (Monte Carlo)")
    xs = np.linspace(0.02, Rmax/Rd*0.999, 400)
    ax[0].semilogy(xs, Sigma_flat(xs*Rd/Rmax)*(Rd/Rmax)**2, c=C3, lw=1.3, label="seed, flat curve (analytic)
        ")
    ax[0].semilogy(xs, np.exp(-xs)/(2*np.pi), c=C2, ls="--", lw=1.5, label=r"exponential, same $R_d$")
    ax[0].axvline(rc/Rd, c="0.6", lw=0.8, ls=":"); ax[0].text(rc/Rd*1.05, 2e-4, r"$r_c$", color="0.4")
    ax[0].axvline(Rmax/Rd, c="0.6", lw=0.8, ls=":"); ax[0].text(Rmax/Rd*0.83, 2e-4, r"edge", color="0.4")
    ax[0].set(xlabel=r"$R/R_d$", ylabel=r"$\Sigma\,R_d^2/M_d$", xlim=(0, 6.5), ylim=(1e-4, 3))
    ax[0].legend(frameon=False, fontsize=7.5, loc="upper right"); ax[0].set_title("(a) surface density",
        fontsize=10)
    # (b) cumulative mass
    xx = np.linspace(0, 6.5, 300); Fs = np.searchsorted(np.sort(R), xx*Rd)/len(R)
    ax[1].plot(xx, Fs, c=C1, lw=1.8, label="seed")
    ax[1].plot(xx, 1-(1+xx)*np.exp(-xx), c=C2, ls="--", lw=1.5, label="exponential")
    ax[1].axhline(0.5, c="0.8", lw=0.6)
    ax[1].set(xlabel=r"$R/R_d$", ylabel=r"$M_d(<R)/M_d$", xlim=(0, 6.5), ylim=(0, 1.02))
```

```
    ax[1].legend(frameon=False, fontsize=8, loc="lower right"); ax[1].set_title("(b) enclosed disc mass",
        fontsize=10)
    # (c) R_d / r_c against lambda
    ll = np.linspace(0.012, 0.11, 200); yy = np.array([y_of(l_) for l_ in ll])
    ax[2].loglog(lams, q_seed, "o", c=C1, ms=5, label=r"seed disc, exponential fit at 1-4 $R_d$")
    ax[2].loglog(lams, q_half, "s", c=C1, mfc="none", ms=5, label=r"seed disc, $R_{1/2}/1.678$")
    ax[2].loglog(ll, np.sqrt(2)*ll/2*(1+1/(4*yy))*yy/(1+1/(4*yy)), c=C2, ls="--", lw=1.4,
                 label=r"Mo et al. (1998): $\lambda y/\sqrt{2}$")
    ax[2].axhline(0.67, c=C3, lw=1.2); ax[2].text(0.0125, 0.69, r"SPARC dwarfs: median 0.67, $\pm1\sigma$
        (0.30 dex)", color=C3, fontsize=7.5)
    ax[2].axhspan(0.67/10**0.30, 0.67*10**0.30, color=C3, alpha=0.12, lw=0)
    from matplotlib.ticker import FixedLocator, NullLocator, FormatStrFormatter
    ax[2].xaxis.set_major_locator(FixedLocator([0.015, 0.02, 0.035, 0.05, 0.07, 0.1])); ax[2].xaxis.
        set_minor_locator(NullLocator())
    ax[2].xaxis.set_major_formatter(FormatStrFormatter("%g"))
    ax[2].yaxis.set_major_locator(FixedLocator([0.3, 0.5, 0.7, 1, 1.5, 2, 3])); ax[2].yaxis.
        set_minor_locator(NullLocator())
    ax[2].yaxis.set_major_formatter(FormatStrFormatter("%g")); ax[2].set_ylim(0.3, 3.2)
    ax[2].set(xlabel=r"spin $\lambda$", ylabel=r"$R_d/r_c$ ($f_j=1$)")
    ax[2].legend(frameon=False, fontsize=7.5, loc="lower right"); ax[2].set_title(r"(c) disc and core ($\
        Lambda=0$)", fontsize=10)
    fig.tight_layout(); fig.savefig("fig_seed_disc.png", dpi=250)
    print("figure written: fig_seed_disc.png")
```

Output (`code/seed_disc_output.txt`):

```
lambda = 0.035, y = 50.5, r_c = 0.0198 R_b, R_max = 0.1492 R_b
exponential fit (1-4 R_d): R_d = 0.0349 R_b = 0.234 R_max; R_d/r_c = 1.76
Mo-Mao-White (j_d = 2 v R_d): R_d = 0.0249 R_b; R_d/r_c = 1.26; ratio 1.402
half-mass radius: 1.265 R_d (exponential 1.678); edge R_max = 4.28 R_d
R_d from the half-mass radius (R_half/1.678) = 0.0263 R_b; R_d/r_c = 1.33; ratio to MMW 1.057
mass in R/R_d = 0-0.1, 0.1-0.25, 0.25-0.5: seed [0.017 0.055 0.1148] exponential [0.0047 0.0218 0.0637]
analytic flat-curve F(S) at S=0.1,0.3,0.6: [0.2332 0.5263 0.8148] Monte Carlo (P18 curve): [0.1529 0.5062
    0.822 ]
R_d/r_c (seed fit 1-4 R_d / seed half-mass / MMW): 0.015:1.25/1.02/0.90 0.02:1.47/1.16/1.06
    0.025:1.62/1.24/1.16 0.035:1.76/1.33/1.26 0.05:1.80/1.35/1.28 0.07:1.73/1.31/1.23 0.1:1.57/1.22/1.13
d ln(R_d/r_c)/d ln lambda (seed, around 0.035): 0.16
figure written: fig_seed_disc.png
```